

Bhadra Biju

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Location: Ernakulam, Kerala

PROFILE

Aspiring AI/ML Engineer with a strong foundation in Machine Learning, Deep Learning, and Generative AI. Experienced in developing predictive models, working with transformer-based architectures, and creating systems that can retrieve, reason, and generate human-like responses from large datasets. Currently taking initial steps into autonomous AI systems and their potential to automate intelligent, multi-step tasks. Passionate about developing intelligent, data-driven solutions to solve real-world problems and support impactful decision-making.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Science and AI:** Exploratory Data Analysis (EDA), Machine Learning, Deep Learning, Natural Language Processing (NLP), Generative AI, Retrieval-Augmented Generation (RAG), Large Language Models (LLM)
- **Frameworks and Tools:** TensorFlow, PyTorch, Keras, YOLO, LangChain, Gemini API
- **Libraries:** NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, Hugging Face
- **Cloud and Platforms:** Amazon Web Services(AWS), Windows, Google Colab, Visual Studio Code
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication, Time Management

EXPERIENCE

- **Data Science Intern** Kakkanad, Kerala
Luminar Technolab (Full-time) June 2025 - May 2026
 - **Data Preprocessing & Feature Engineering:** Cleaned, transformed, and prepared datasets using Python (Pandas, NumPy), including feature scaling and encoding to improve model performance.
 - **Exploratory Data Analysis (EDA):** Performed EDA to identify trends, patterns, and correlations; derived insights to guide model selection and training strategies.
 - **Machine Learning Model Development:** Built, trained, and evaluated machine learning models using Scikit-learn, applying appropriate metrics to assess performance.
 - **Deep Learning Implementation:** Developed deep learning models for pattern recognition tasks using neural networks and optimized them through iterative experimentation.

PROJECTS

- **YouTube Video Knowledge Base (RAG, Generative AI):** Developed a RAG-based QA system that transcribes YouTube playlists using Whisper, stores semantically chunked transcripts in a FAISS vector database, and retrieves context to generate grounded answers with clickable timestamp citations using a large language model. **Tech:** Python, Whisper, yt-dlp, LangChain, FAISS, Sentence Transformers, Gemini API, Streamlit.
- **AI Karate Trainer for Hand Posture (Deep Learning, Computer Vision):** Developed a deep learning-based system to recognize karate hand symbols and correct hand position and posture in real time, functioning as an AI trainer. Utilized hand landmark detection to analyze movements and provide accurate feedback. **Tech:** Python, MediaPipe, OpenCV, NumPy, Hand Landmarks.
- **Fraudulent Job Offer Letter Detection (NLP, Machine Learning):** Implemented an NLP-based solution to identify fraudulent job offer letters using text preprocessing and feature extraction techniques. Deployed an interactive Streamlit interface to display real-time prediction results. **Tech:** Python, Pandas, NumPy, Matplotlib, Seaborn, NLP, Scikit-learn, Streamlit.

EDUCATION

- **University of Calicut** Thrissur, Kerala, India
Bachelor of Computer Applications - BCA; GPA: 7.00 August 2022 - May 2025
Courses: Database Management Systems, Data Structures, Computer Graphics, Computer-Oriented Numerical and Statistical Methods, Web Programming, Python.
- **Amrita Vidyalayam** Ernakulam, Kerala, India
Commerce with Informatic Practices ; Percentage: 85 July 2020 - May 2022
Courses: Python, SQL, Data Structures, Web Development, Accounting, Business Studies, Web Programming.

CERTIFICATIONS

- NACTET Certified in Python Data Science AI ML - Luminar Technolab
- Python Programming - Data Analysis and Application Design with Project conducted by ICT Academy of Kerala
- AWS for Beginners – Introduction to cloud concepts and AWS services from Simplilearn.